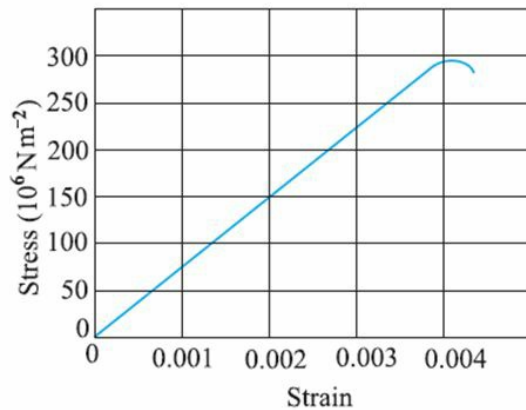
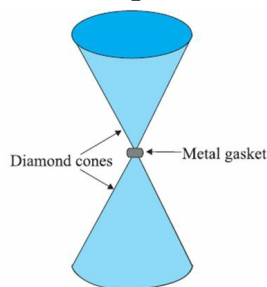


CBSE Test Paper 04
Chapter 9 Mechanical Properties of Solids

1. Figure shows the strain-stress curve for a given material. What is the approximate yield strength for this material? **1**



- a. $3.6 \times 10^8 \text{ N/m}^2$
b. $3.2 \times 10^8 \text{ N/m}^2$
c. $3.4 \times 10^8 \text{ N/m}^2$
d. $3.0 \times 10^8 \text{ N/m}^2$
2. Anvils made of single crystals of diamond, with the shape as shown in Figure, are used to investigate behaviour of materials under very high pressures. Flat faces at the narrow end of the anvil have a diameter of 0.50 mm, and the wide ends are subjected to a compression force of 50,000 N. What is the pressure at the tip of the anvil? **1**



- a. $2.1 \times 10^{11} \text{ Pa}$
b. $2.5 \times 10^{11} \text{ Pa}$
c. $2.9 \times 10^{11} \text{ Pa}$
d. $3.2 \times 10^{11} \text{ Pa}$
3. Rectangular section is rarely used in beams because **1**

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- a. less material in kilos is required
 - b. more material in kilos is required
 - c. stresses are uniform in a rectangular section
 - d. stresses are always plastic in rectangular section

4. Stress is **1**

- a. total applied force
- b. force per unit length
- c. restoring force per unit area
- d. three point average of forces

5. You hang a flood lamp from the end of a vertical steel wire. The flood lamp stretches the wire 0.18 mm and the stress is proportional to the strain. How much would it have stretched if the wire were twice as long? **1**

- a. 0.36 mm
- b. 0.34 mm
- c. 0.40 mm
- d. 0.38 mm

6. How is compressibility related to bulk modulus? **1**

7. A thick wire is suspended from a rigid support, but no load is attached to its free end. Is this wire under stress? **1**

8. What does the slope of stress versus strain graph indicate? **1**

9. i. Draw stream lines for a clockwise spinning sphere.
ii. Derive equation of continuity. **2**

10. The Young's modulus of a wire of length L and radius r is Y . If the length is reduced $\frac{L}{2}$ to and radius $\frac{r}{4}$, what will be its Young's modulus? Why? **2**

11. A wire of length 2.5 m has a percentage strain of 0.012% under a tensile force. Determine the extension in the wire. **2**

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12. How is the knowledge of elasticity useful in selecting metal ropes used in cranes for lifting heavy loads? **3**
13. Establish the relation $S_{nth} = u + \frac{a}{2}(2n - 1)$ where the letters have their usual meanings. **3**
14. A particle is moving along a straight line and its position is given by the relation $x = (t^3 - 6t^2 - 15t + 40)\text{m}$ Find: **3**
- The time at which velocity is zero.
 - Position and displacement of the particle at that point.
 - Acceleration of the particle at that point.
15. Determine the volume contraction of a solid copper cube, 10 cm on an edge, when subjected to a hydraulic pressure of $7.0 \times 10^6 \text{ Pa}$. **5**
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